



Pair Trading Analysis of ETF and its Constituent Stock

This presentation explores pair trading of an ETF and one of its underlying stocks. We use quantitative techniques and advanced data analysis to uncover trading insights and risk management strategies.

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a 31 dez. 2024 ▼	
Ticker	Nome ⓘ
FSLR	FIRST SOLAR INC
IBE	IBERDROLA SA
SSE	SSE PLC
ENPH	ENPHASE ENERGY INC
VWS	VESTAS WIND SYSTEMS
600900	CHINA YANGTZE POWER LTD A
9502	CHUBU ELECTRIC POWER INC
EDP	EDP ENERGIAS DE PORTUGAL SA
SUZLON	SUZLON ENERGY LTD
NXT	NEXTRACKER INC CLASS A
ORSTED	ORSTED
EQTL3	EQUATORIAL SA
ELET3	CENTRAIS ELETR BRAS-ELETRC
BREN	BARITO RENEWABLES ENERG

ETF Context and Analytical Methods

ETF — iShares Global Clean Energy Transition UCITS (IQQH.DE)

We focused on a major energy sector ETF, representative of a large market segment.

Data from 2023-01 to 2025-01

- ### Data Techniques
- Principal Component Analysis (PCA) for dimensionality reduction
 - DBSCAN clustering based on correlation matrices

Correlation Matrix Overview

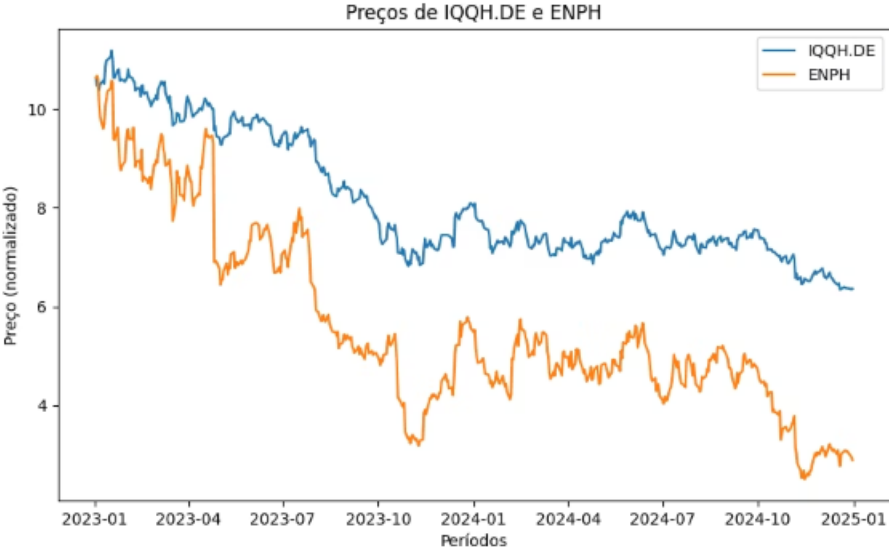
Ticker	Valor	Ticker	Valor
ENPH	0.96	600905.SS	0.87
NPI.TO	0.92	EDP.LS	0.80
ORSTED.CO	0.91	ORA	0.77
601012.SS	0.90	BLX.TO	0.74
EDPR.LS	0.89	VWS.CO	0.63

This correlation matrix forms the basis for clustering and selecting candidate pairs.

Best Pair Selection and Visual Analysis

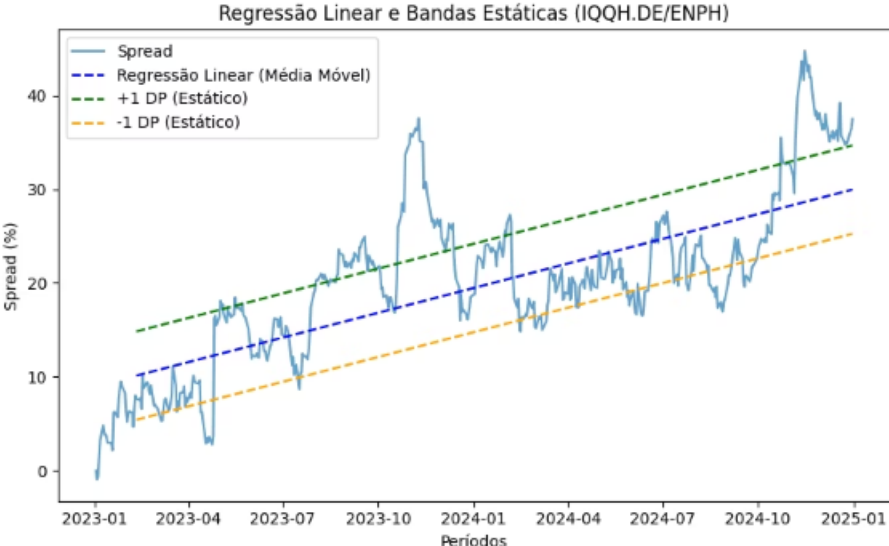
1

Prices Chart



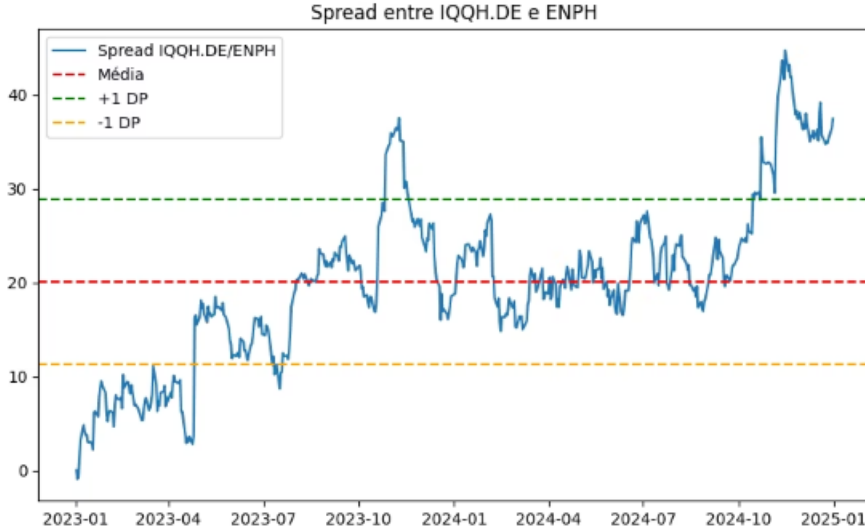
2

Stationary Chart



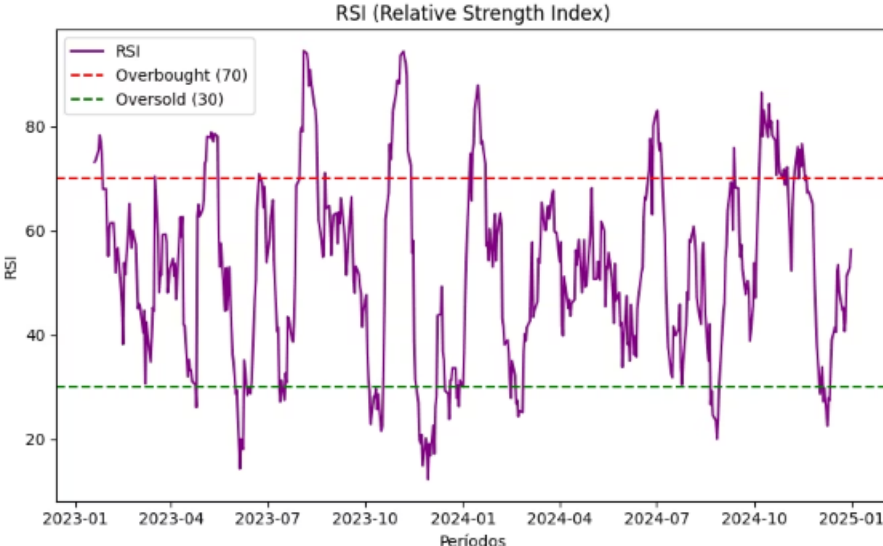
3

Linear Regression Chart



4

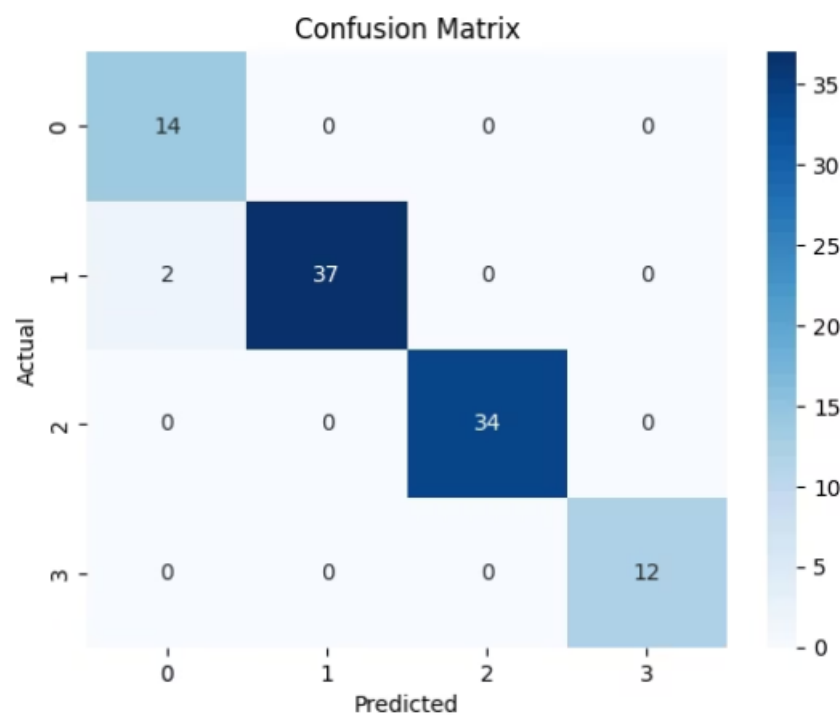
RSI Chart



Random Forest Indicator Implementations

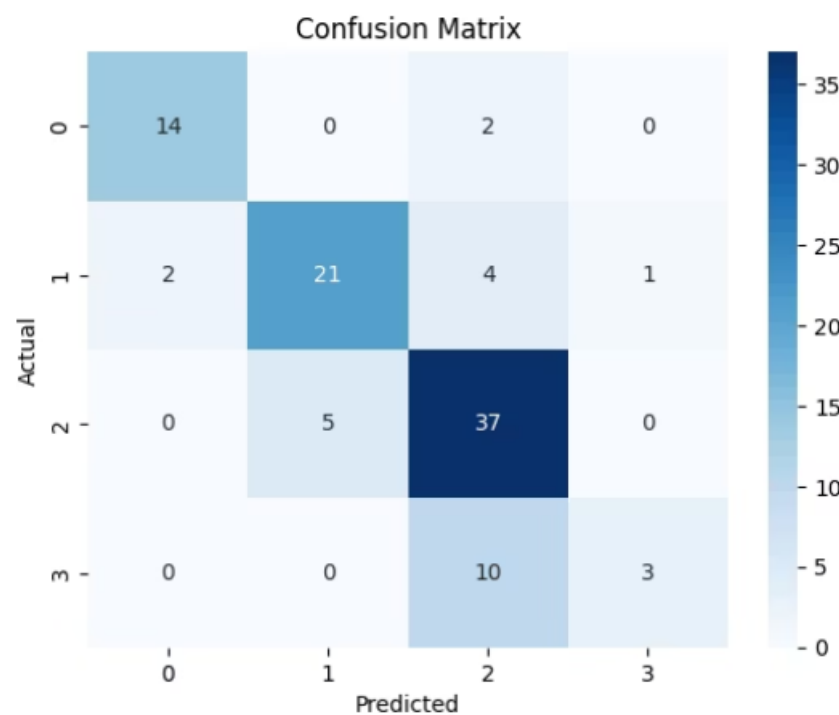
Stationary Model

Uses raw spread data for feature extraction and prediction.



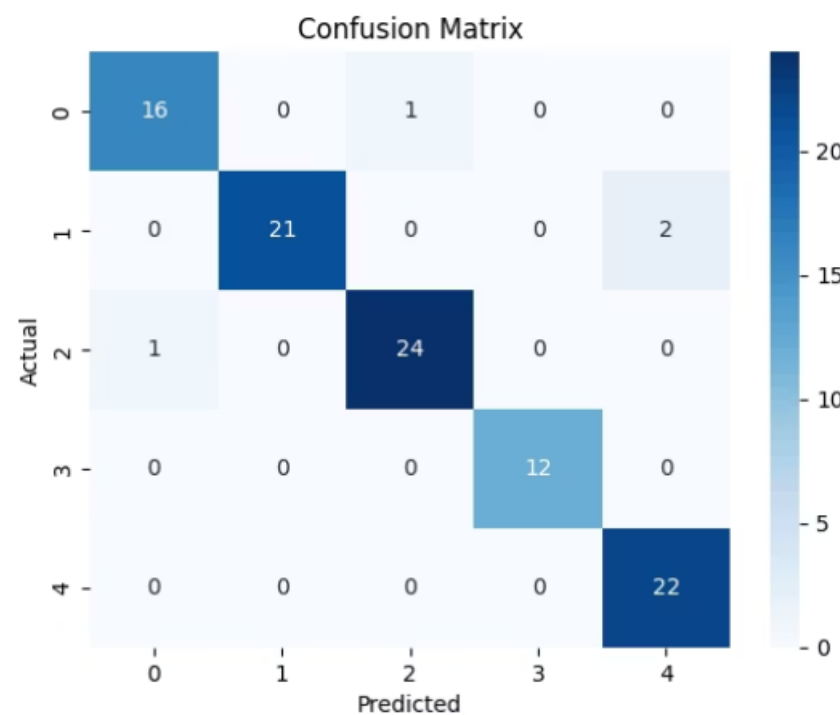
Linear Regression Model

Incorporates moving averages of spread to capture trends over time.



RSI-based Model

Leverages momentum features from the Relative Strength Index.



Feature importance varies across implementations, highlighting different market aspects. Using different indicators to help predict the model, such as the spread, z-score, spread_ma, spread_std, rsi, rolling_zscore, seasonal_zscore, diff_zscore, modified_zscore, ewma_zscore.

Backtesting Methodology

Testing Strategies

Backtest models using Stationarity, Linear Regression and RSI approaches .

Model Evaluation

Performance metrics guided iterative improvements.

Stop Loss

Used to minimize the losses based on the previous records.



Dynamic Stop Loss Implementation

Stop Loss Concept

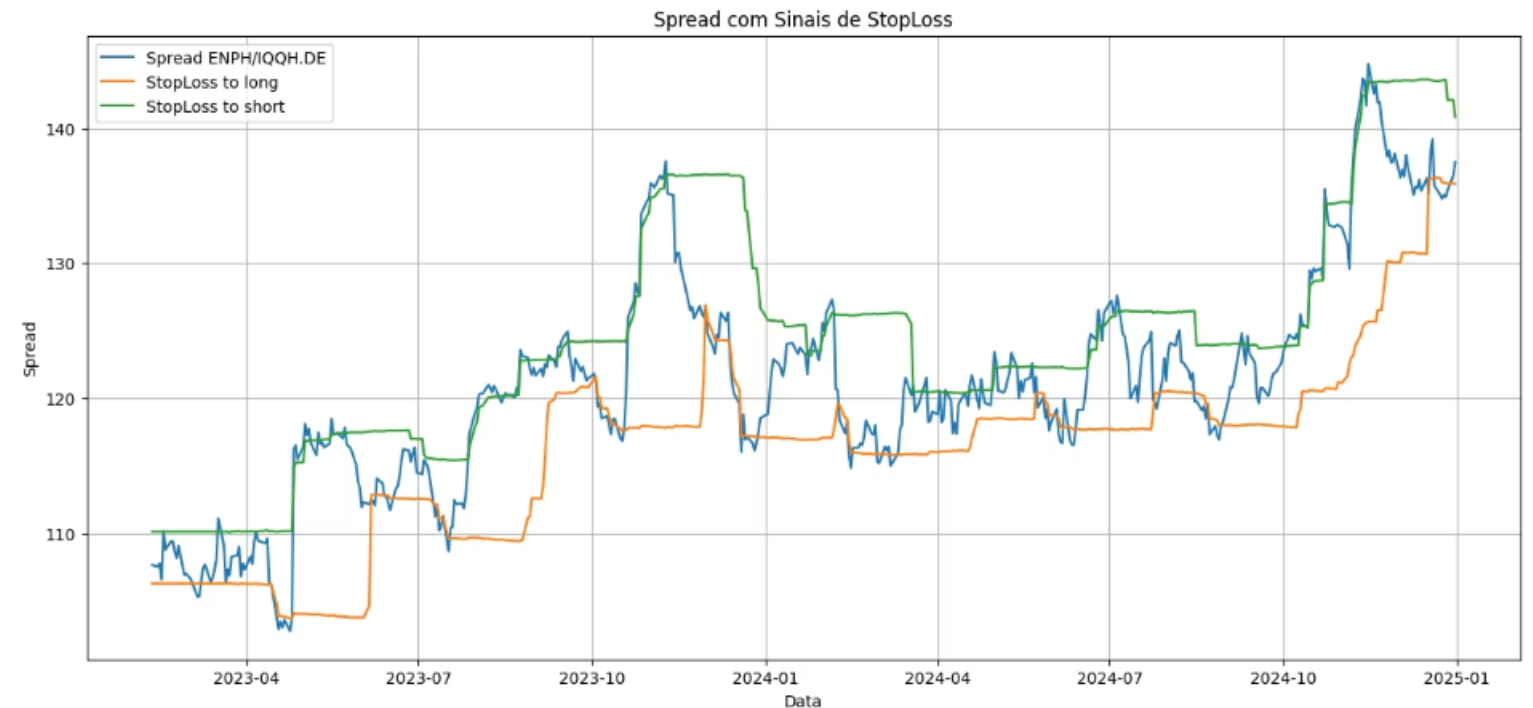
A risk management tool that limits potential losses on positions.

Importance

Minimizes downside risks and protects capital during adverse market moves.

Dynamic Aspect

Our Stop Loss is dynamically altering based on the local minima of the last 30 days.

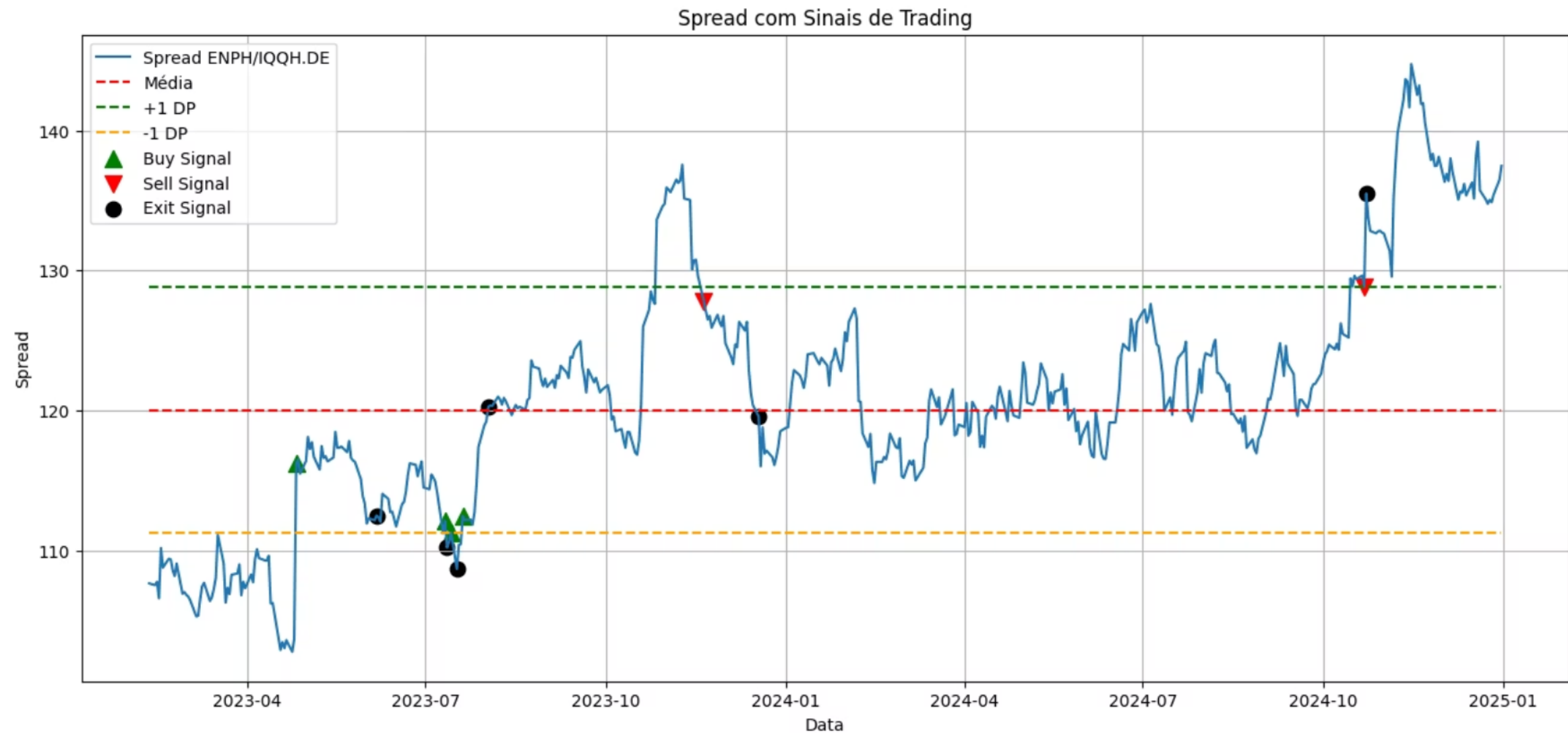


The Stop Loss is dynamically calculated not only for a long scenario but also in case of the investor finding a short opening.

Stop Loss Results – Stationary Spread

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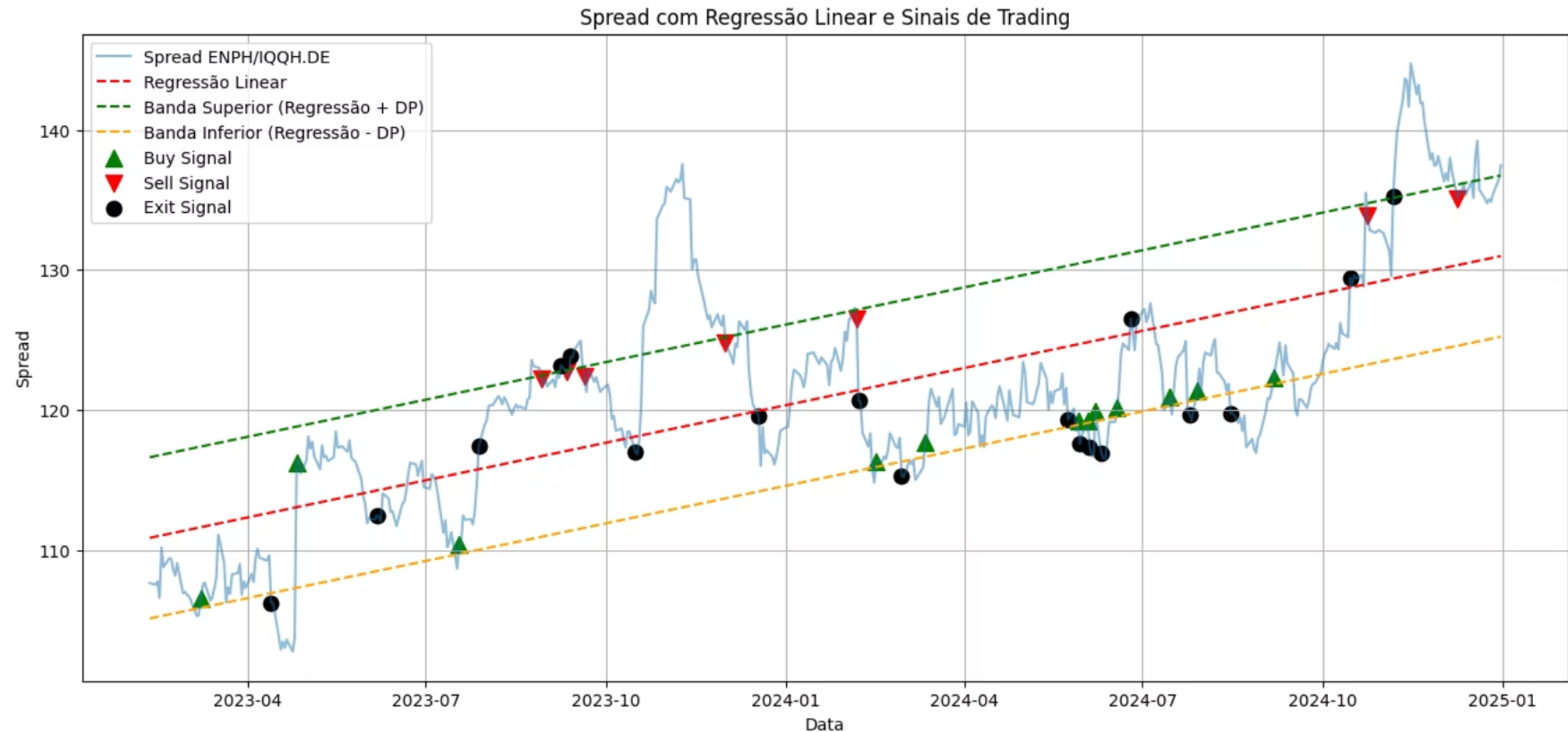
In the Stationary Spread of our statistically best pair, we have 6 openings finding a Stop Loss example around 2023-08.



Stop Loss Results - Linear Regression

ENPH & IQQH.DE

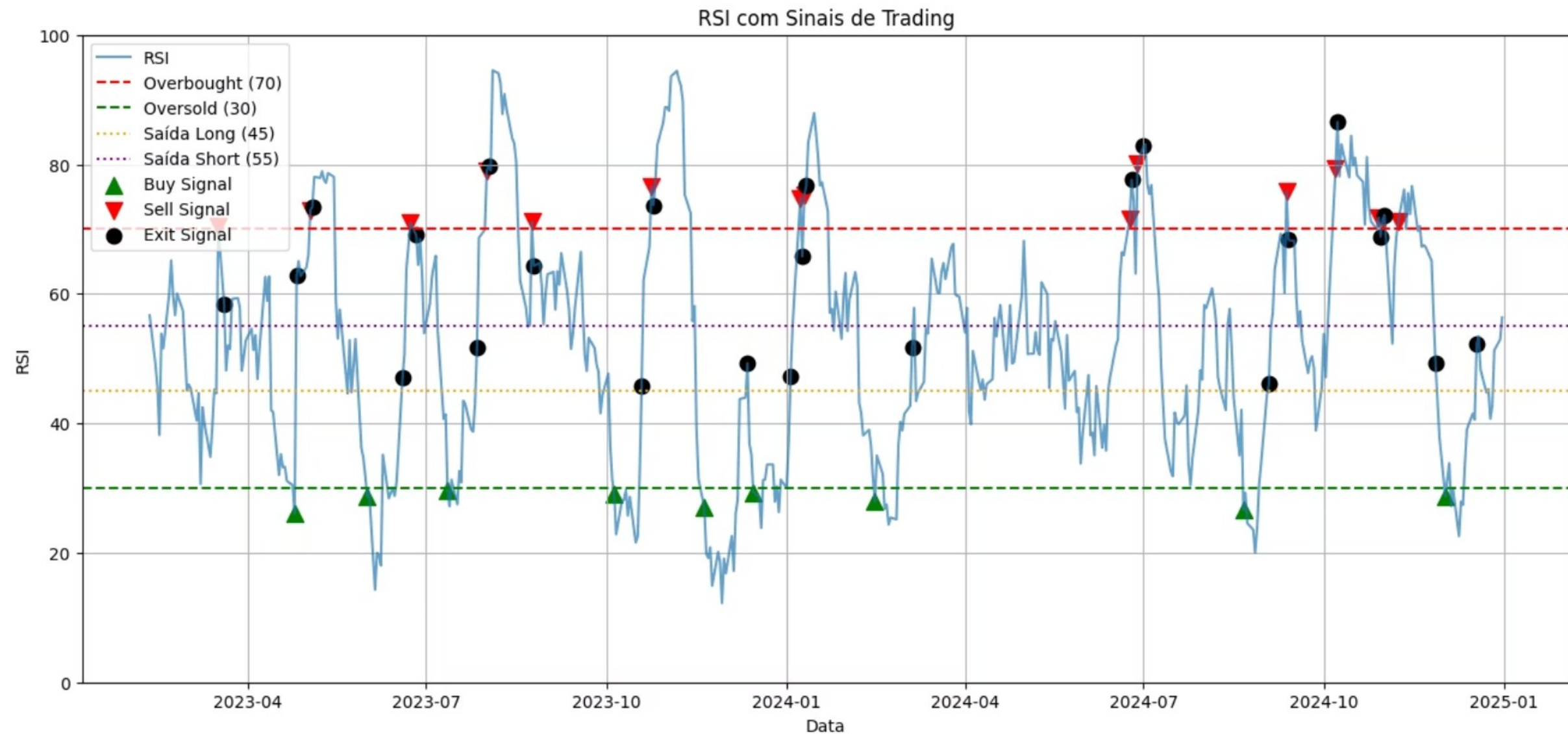
In the Stationary Spread of our statistically best pair, we have 19 openings finding a Stop Loss example around 2023-09.



Stop Loss Results - RSI

ENPH & IQQH.DE

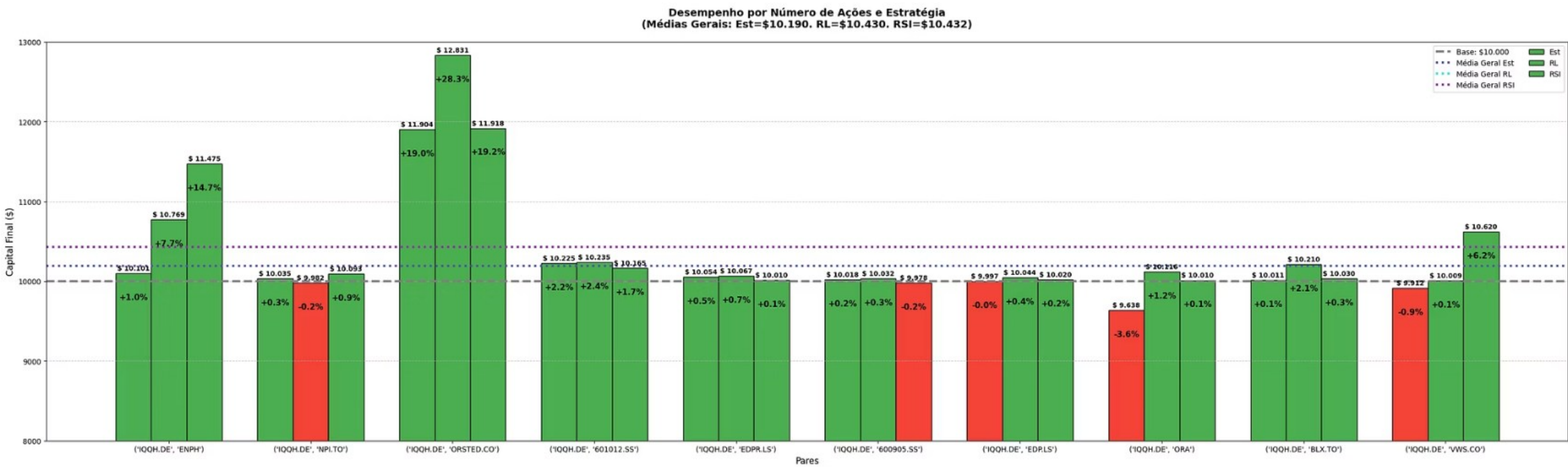
In the Stationary Spread of our statistically best pair, we have 24 openings, finding several stop loss cases.



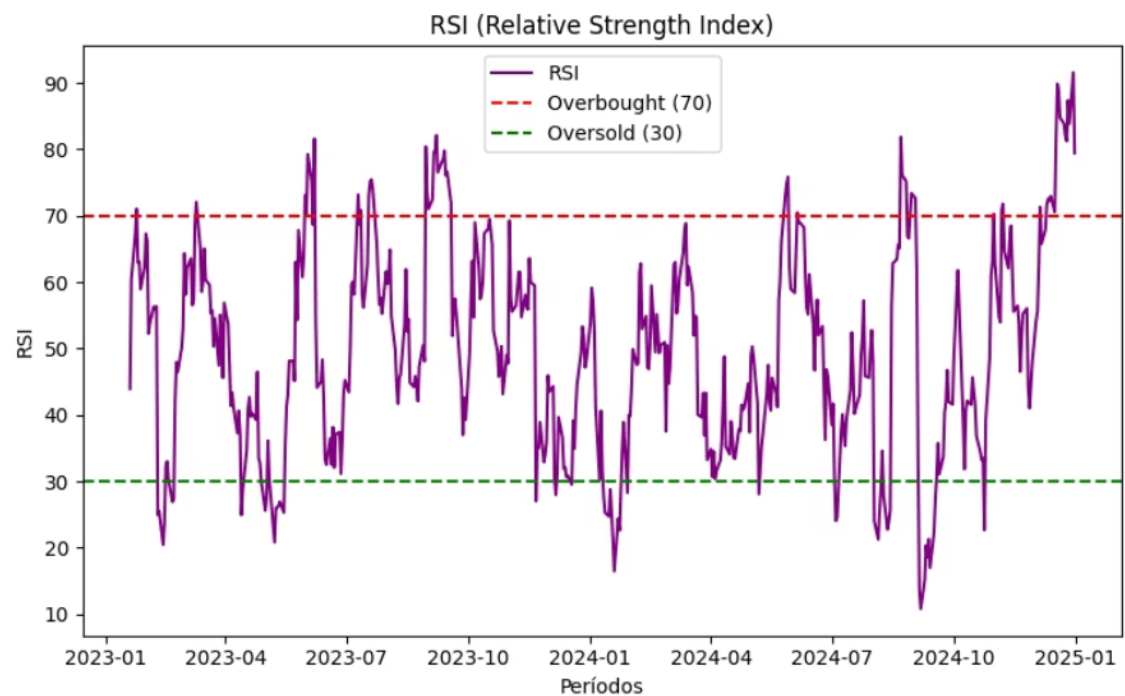
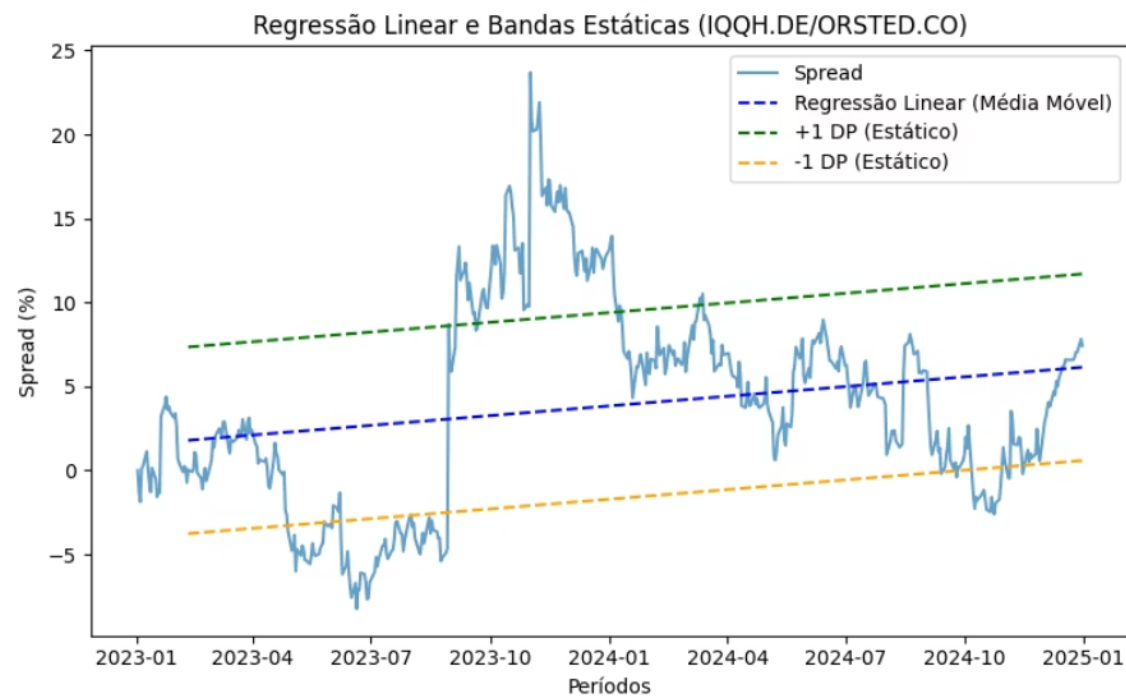
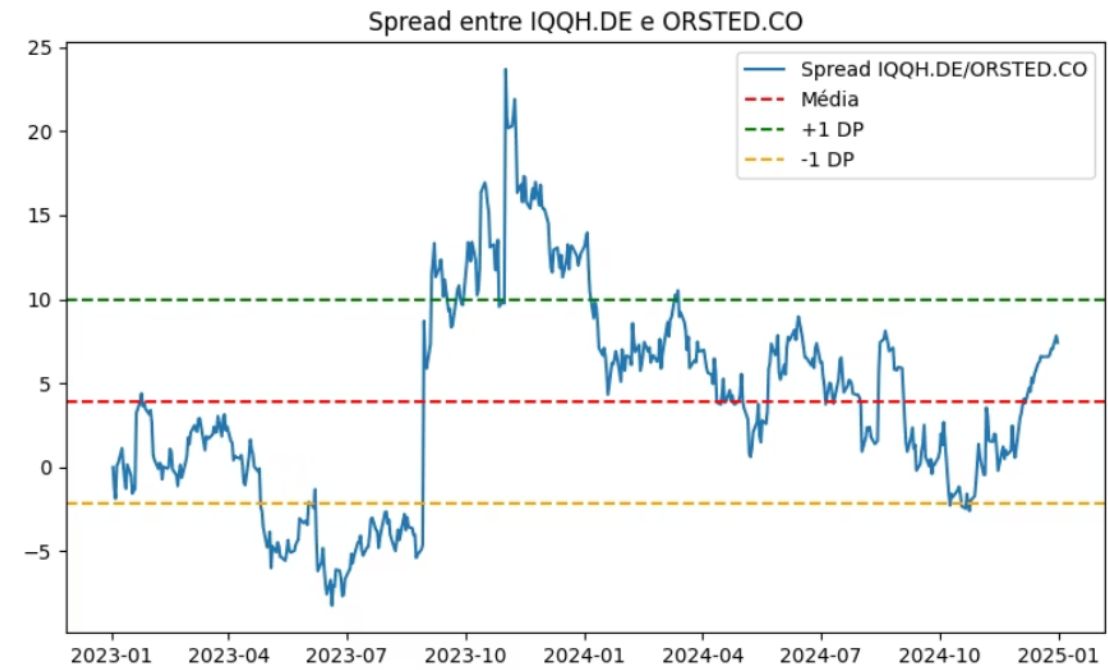
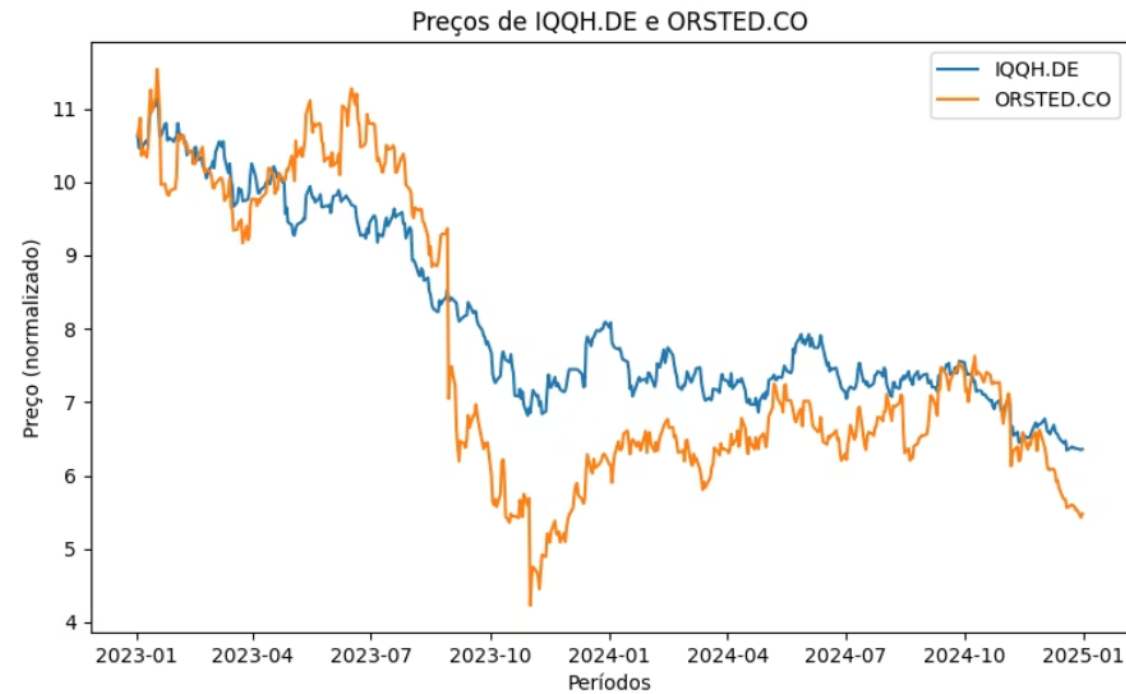
Overall Stats for each Pair

Surprisingly, the 3rd pair has the best score by a large margin using our Linear Regression model.

However, we can also note that in our best pair (1st) we can find that RSI provides the best gains.



Spreads of the 3rd Pair



Summary and Next Steps

This approach combines innovative analysis while mitigating eventual losses with Dynamic Stop Loss.

Refine Random Forest Features

Focus on enhancing indicator selection for robust prediction.

Value Prediction

Implement a model that predicts the value of the next day so to mitigate losses even further.

Real-time Testing

Deploy models live with active risk controls.



The End

Questions?

